Design → Demo → Deploy

An AI intake & referral agent for Workers' Comp DME — live today, deployable in your environment in 8–12 weeks. A hands-on workshop to prove the design, the demo, and the rollout.

1 Day	Agent 1	$0.032	24
WORKSHOP	LIVE NOW	PER REFERRAL	CMS RULES

Make intake & referral management an advantage

Today, DME referral intake is manual, slow, and error-prone — staff rekey faxes, chase missing fields, and miscode claims. The agent turns it into autonomous, clean, route-ready episodes — with a human in the loop at every flagged exception.

The cost of manual intake

Hours of rekeying per referral · coding errors that trigger denials · missing auth refs that stall episodes · no audit trail · can't scale with volume.

The agent advantage

Reads every page, extracts 25 fields, validates against CMS rules, chases gaps by email & voice, and routes a clean episode — in seconds, at $0.032 each, fully audited.

The principle: autonomous where it's confident, human where it isn't. The agent never guesses on a clinical or coding decision — it escalates.

What you'll see — on real referrals

Not a mock-up. A real Claude agent processes actual referral PDFs end-to-end, on screen, on your representative data.

Referral arrives
Fax, voice or portal

→

Extracts 25 fields
Reads every page

→

Resolves ICD conflict
Catches miscodes

→

Chases the gap
Email + voice

→

Routes episode
Clean & audited

Watch for #1

The agent flags an ICD conflict (M23.611 vs S83.209A) and explains *why*, citing the document.

Watch for #2

A missing auth ref is caught as a hard block — the episode is held, outreach is drafted to the adjuster by name.

Try it yourself

Scan the QR → open the live demo on your phone → run a referral and try to "stump the agent."

The big picture — six layers, one pipeline

Every referral, regardless of channel, flows through the same six-layer pipeline. AI does the reading and reasoning; a deterministic rule engine does the compliance; a human decides the close calls.

Multi-Channel Intake
Fax · Voice · Portal

→

AI Extraction
Large Language Model

→

OWL Knowledge Graph
Deterministic rules

→

Completeness Check
Gap classification

→

Human Gate
<80% confidence

→

Route Output
Locked + dispatched

AI where judgement helps

The LLM reads unstructured documents, extracts structured fields, and reasons about conflicting evidence across pages.

Rules where certainty is required

Coding compliance is never left to probability — the knowledge graph confirms or rejects with a CMS citation. Auditable by construction.

How the agent works — step by step

1. **Channel Intake**
   Ingests fax PDFs (counts every page) or answers the live voice line — 14 fields captured in ~107 seconds. Both channels feed one queue.

2. **AI Extraction — 25 structured fields**
   Reads all documents at once: patient, claim, ICD-10, HCPCS, physician NPI, adjuster, address, auth ref. Cross-references documents and scores its own confidence 0–100%.

3. **OWL Knowledge Graph — 24 CMS rules**
   Every code validated against deterministic rules sourced from CMS guidelines. Confirms, rejects, flags or escalates — each with its citation. Runs in <150 ms.

4. **Completeness Check**
   Classifies gaps by criticality: hard blocks (auth ref, appt window) hold the episode; required gaps (transport, language) collected in parallel; clean episodes route untouched.

5. **Human Review Gate**
   Below 80% confidence the agent stops — routes to a specialist with full reasoning, source docs, one-click approve/override, logged with reviewer identity.

6. **Route Output**
   Locks the 25-field episode record, saves the full audit trail, triggers vendor dispatch with the right code, equipment and patient.

Multi-channel in, autonomous outreach out

The agent meets referrals where they are — and chases what's missing without a human lifting a finger.

Fax

Ingests all PDFs — referral form, clinical notes, prescription — and counts every page before forming an opinion.

Voice (inbound)

Answers the intake line 24/7, runs a structured conversation, captures 14 fields live — 107s average.

Portal upload

Accepts portal submissions into the same queue — same process, same output.

Email outreach

Drafts a specific email to the adjuster *by name* — lists exact missing fields, cites the claim number, sets a 24-hour deadline. Logged on send.

Outbound voice call

No reply in 24h? The agent calls — 84 seconds — confirms auth ref, collects the appointment window, updates the episode from the transcript.

Four-layer defense against hallucination

The single most important design choice: the agent is built so it cannot quietly guess on a clinical or coding decision.

1 · Document cross-reference

Compares ICD codes across every document — mismatches flagged with document-level reasoning.

2 · KG validation

Deterministic CMS rules confirm or reject each code — no probability on coding.

3 · Confidence threshold

The agent scores itself 0–100%. Below 80%, it does not proceed.

4 · Human gate

Low-confidence or conflicting cases route to a specialist for a binding decision.

Status priority — the guardrail that makes it deployable: REJECTED > ESCALATED > FLAGGED > VALIDATED. The safest outcome always wins — the agent never marks an episode “clean” while a live coding concern exists.

Every decision is logged & reviewable

When the agent escalates, the specialist sees *why* — not just a flag. When it acts, the reasoning survives audit.

The human review queue

- ICD conflicts below 80% → specialist queue, episode blocked from dispatch
- Reviewer sees the agent's full reasoning + source documents
- Approve or override in one click
- Decision logged with reviewer identity + timestamp

The audit trail (per episode)

- 25 structured fields + KG validation results
- Every rule that fired, with CMS citation
- ICD conflicts and how they were resolved
- Email body + call transcript
- Every confidence score, every decision

> *"When confidence falls below 80%, the agent stops. It doesn't guess on clinical decisions — it routes to a specialist with the full conflict detail."*

A deterministic rule engine — cited, not guessed

24 rules extracted from CMS ICD-10-CM Official Guidelines FY2026 and LCD/NCD documents. Every validation is deterministic, cited, and auditable — no AI probability on coding decisions.

24	7	100%	<150ms
RULES	CATEGORIES	DETERMINISTIC	PER EPISODE

Why OWL, not a flat rule list

OWL (W3C Web Ontology Language) enforces class hierarchies, property chains and disjointness — so the graph stays logically consistent as rules are added. A reasoner can infer new facts automatically.

Built from the source of truth

CMS ICD-10-CM FY2026 (144 pages) + LCD L33803 (walkers/rollators) + NCDs → structured into `rules.json`, each rule citing its exact CMS section.

Seven rule domains · four possible actions

Post-Surgical Coding

6 rules — enforce S-codes (injury) over M-codes (degenerative) after surgery.

Pain Classification

3 rules — postprocedural G89.18 vs trauma G89.11; sequencing.

Laterality

2 rules — right/left mismatches across documents; escalate conflicts.

HCPCS Coverage

4 rules — confirm equipment codes covered for the diagnosis.

Equipment Spec

2 rules — bariatric thresholds (E0143 300lb vs E0149 400lb).

Workers Comp

3 rules — required WC fields; reject M-codes as primary on injury claims.

Authorization / Docs

4 rules — missing auth ref; 5-element medical-necessity check.

Every rule cites CMS

e.g. `§I.C.19.a`, `LCD L33803` — traceable to source.

CONFIRM

Code validated — contributes to VALIDATED status.

REJECT

Coding error identified — replacement + source given.

FLAG

Concern needs review — may not block dispatch.

ESCALATE

Conflicting evidence — human decision required.

Live example — the Holloway referral

James Holloway · ACL reconstruction · E0143 rollator · 285 lbs. The KG ran 24 rules in <100 ms — 5 fired.

DME-ICD-001 — REJECT
M23.611 (degenerative) on a post-surgical episode. Replace with S83.209A. Source §I.C.19.a.

DME-ICD-002 — CONFIRM
S83.209A correct post-ACL encounter code (7th char 'A'). Confidence 100%.

DME-ICD-007 — CONFIRM
E0143 rollator covered for S83.209A per LCD L33803.

DME-ICD-023 — FLAG
285 lbs — within 15 lbs of the 300 lb limit. Confirm bariatric spec before dispatch. Catches a costly wrong-equipment return.

DME-ICD-015 — FLAG
WC claim, auth ref blank — blocks billing. Triggers outreach to adjuster Linda Torres.

Final status: FLAGGED — 1 rejection (resolved by LLM self-correction), 2 confirmations, 2 flags. Full citation trail attached to the route output.

Fits your stack — no rip-and-replace

The agent connects to what you already run, three ways, and is built to slot into the wider operations chain.

API

Direct integration with your case-management / EHR / intake portal where APIs exist.

RPA

Robotic automation drives legacy systems that have no API — the agent operates the screens.

File

Batch file exchange (fax gateway, SFTP drops) for the simplest, lowest-touch start.

Connects to

Fax gateway · intake portal · case-management system · scheduling · billing · EHR — inbound and outbound.

Carrier rules live in the graph

Payer-specific rules are added to the knowledge graph as config — no code changes to support a new carrier.

PHI stays inside your boundary

Healthcare-grade from day one. This is the question your technology leader asks first — here is the answer.

HIPAA + BAA

HIPAA-compliant deployment. LLM-provider BAA available. PHI never leaves your boundary.

Access & audit

Role-based access, full audit logging of every decision, configurable data-retention policy.

Your environment

Containerized Python — runs on your infrastructure, your cloud, or ours. You receive source, logs, runbooks.

Bring this to the security review: data-boundary diagram, HIPAA/PHI checklist, draft BAA summary, retention policy — all on the table at the workshop.

Enterprise-ready operations

Everything a technology leader needs to say yes — and to run it confidently after go-live.

Monitoring

Live system health, AI quality metrics, throughput & cost dashboards.

Support

Dedicated support through pilot and go-live; runbooks for your team.

Scales

Demonstrated 40 → 1,000 → 3M referrals on the same architecture.

Improves

Learns your payers and patterns; review-queue feedback feeds the next cycle.

> *"Your team receives full source code, audit logs, and runbooks — with dedicated support through pilot and go-live."*

The economics — measured, not estimated

Real token consumption from 40 live referrals, extrapolated to production volume.

$0.032	40	1,000	3M
PER REFERRAL	LIVE REFERRALS MEASURED	EXTRAPOLATED	AT FULL SCALE

What drives the cost

A few cents of LLM tokens per referral for extraction + reasoning. The knowledge graph runs locally — no per-call cost, no network round-trip.

Why it stays flat at scale

Cost is per-referral and linear — no per-seat licensing, no overtime, no rekeying. Volume spikes don't spike headcount.

Where the return comes from

We'll plug your monthly volume into the calculator live — these are the levers.

Time saved

Seconds of agent processing replace minutes-to-hours of manual rekeying and gap-chasing per referral.

Fewer denials

Coding errors caught before submission — M-code rejections, missing auth refs — mean fewer reworks and denials.

Throughput

Process peak volume without adding staff; clean episodes route with zero human touch.

The pitch in one line: at $0.032 per referral, the cost of the agent is a rounding error against a single avoided denial or hour of staff time. We'll prove the payback on *your* numbers.

What we built · how we deploy · how it improves

01 — What we built

A modular agent network: AI extraction, OWL knowledge graph (24 CMS rules), four-layer defense, email + voice outreach, full audit trail. **75 referrals in 60 seconds at $0.032 each.**

02 — How we deploy it

Containerized Python in your environment. API / RPA / file integration — no rip-and-replace. BAA + HIPAA config. **8–12 weeks from kickoff to go-live.**

03 — How it gets smarter

Tunes to your payers and patterns; KG extended with your carrier rules (config, not code); confidence calibrated to your tolerance. **Improves from month 3+.**

> *“We can run a 48-hour proof-of-concept on a sample of your actual intake packages — no integration required, no PHI shared.”*

8–12 weeks from kickoff to go-live

A staged, low-risk path — shadow-mode pilot before anything goes live.

1 WEEKS 1–2
Integration Discovery
Fax/portal mapping
Case-mgmt API review
PHI boundary design
Payer-rule inventory

2 WEEKS 3–5
Environment Setup
Infra provisioning
API connectivity
BAA + HIPAA config
KG customization

3 WEEKS 6–8
Test & Validate
Real referral testing
ICD accuracy review
Escalation tuning
Staff training

4 WEEKS 9–12
Pilot & Go-Live
Shadow-mode pilot
Live processing
Monitoring + SLAs
Handoff + runbooks

Anchored to June 10: Discovery starts the week of June 16 → go-live by early September.

A platform, not a point solution

Referral to payment — an 8-agent value chain. Agent 1 is live; each agent is independently deployable in your environment.

Intake & Referral • LIVE	**Scheduling** NEXT	**Documentation**	**Billing**	**Collections**	**Cash Apps**	**Appeals**	**Financial Shared Svcs**

Start where the pain is

Intake is live and proven. Scheduling is the natural next agent — but your priority shapes the order.

One architecture, many agents

Each agent shares the same auditable, human-in-the-loop design — deploy them one at a time, de-risked.

Walk in with a demo. Walk out with a plan.

Before we leave the room, we agree:

- **Pilot scope** — which referral types, which volume, shadow-mode start
- **Success metrics** — % auto-processed, accuracy, handling time, cost/referral
- **Owners & dates** — named owner and date next to every action
- **Security review** — HIPAA/BAA kicked off the week of June 16